Benchmark: **MA.7.GR.2.2** · Real-world cylinder surface area Student code: _____ Calculator: yes

Learning goal: I can solve problems involving surface area of cylinders. (circle your level after scoring)

4	I can solve a multi-step surface-area problem, such as a label or an open-top can.
3	I can solve a real-world problem with the surface area of a cylinder.
2	I can find the area of a circle and a rectangle.
1	I need help to start this benchmark.

1. Surface area of a right circular cylinder: $\text{SA} = 2\pi r^2 + 2\pi rh$

A cylindrical water tank has a radius of 9 feet and a height of 17 feet. Its whole outside surface, including the top and the bottom, will be covered with sheet metal.

What is the area that the sheet metal must cover? Write your answer in terms of π.

2. Surface area of a right circular cylinder: $\text{SA} = 2\pi r^2 + 2\pi rh$

A cylindrical container is made of plastic. It has a radius of 3 inches and a height of 14 inches, and it has a bottom but no top.

What is the area of the plastic needed to make the container? Write your answer in terms of π.

Turn over →

Benchmark: **MA.7.GR.2.2** · Real-world cylinder surface area Student code: _____

3. Surface area of a right circular cylinder: $\text{SA} = 2\pi r^2 + 2\pi rh$

A cylindrical gift tin with a lid has a radius of 3 inches and a height of 4 inches. The whole tin, including the lid and the bottom, is covered with decorative paper.

What is the area the paper covers, in square inches?

A. 42π

B. 30π

C. 33π

D. 24π

4. Surface area of a right circular cylinder: $\text{SA} = 2\pi r^2 + 2\pi rh$

A cylindrical oatmeal container has a radius of 5 inches and a height of 11 inches. A label wraps all the way around its curved side, with no overlap; the top and the bottom are not covered.

What is the area of the label? Round your answer to the nearest hundredth. Use the π key on your calculator.

5. Surface area of a right circular cylinder: $\text{SA} = 2\pi r^2 + 2\pi rh$

A cylindrical water tank stands on the ground. It has a radius of 7 feet and a height of 17 feet. The curved side and the top will be painted, but not the bottom. One gallon of paint covers 400 square feet, and paint is sold only in whole gallons.

How many gallons of paint must be bought? Use the π key on your calculator.

Score: ____ / 5 **My level: 1 2 3 4** **→ copy both to your progress strip**

Answer key — C-GR.2.2 · MA.7.GR.2.2 · Real-world cylinder surface area

Unit 12: Surface Area and Volume. Regular = on-benchmark item; Rigor = the item that needs the benchmark's reasoning. Mastery on the weekly check = 4 of 5.

Weekly check (5 items)

1. Regular **468π square feet**

The sheet metal covers both bases and the curved surface: the whole surface area.

$\text{SA} = 2\pi(9)^2 + 2\pi(9)(17) = 162\pi + 306\pi = 468\pi$ square feet.

2. Regular **93π square inches**

With no top, the container has one base and the curved surface.

One base: $\pi(3)^2 = 9\pi$.

Curved surface: $2\pi(3)(14) = 84\pi$.

Total: $9\pi + 84\pi = 93\pi$ square inches.

3. Regular **A.** 42π

The paper covers both circular ends and the curved side: the whole surface area.

$\text{SA} = 2\pi(3)^2 + 2\pi(3)(4) = 18\pi + 24\pi = 42\pi$.

4. Regular **345.58 square inches**

The label covers only the curved side, the $2\pi rh$ part of the formula; the $2\pi r^2$ part (the top and the bottom) is not included: $2\pi(5)(11) = 110\pi$.

With the π key: $110 \times \pi \approx 345.58$ square inches.

5. Rigor **3 gallons**

The painted surface is one base and the curved surface: $\pi(7)^2 + 2\pi(7)(17) = 49\pi + 238\pi = 287\pi$ square feet.

$287\pi \approx 901.64$ square feet.

$901.64 \div 400 \approx 2.25$, so 2 gallons would not be enough.

3 gallons must be bought.

Re-check strip (3 items, C-GR.2.2-R)

1. Regular **455.53 square inches**

The planter has one circular base and the curved side, so only one πr^2 is used: $\pi(5)^2 + 2\pi(5)(12) = 25\pi + 120\pi = 145\pi$.

With the π key: $145 \times \pi \approx 455.53$ square inches.

2. Regular **112π square centimeters**

The label covers only the curved side, the $2\pi rh$ part of the formula; the $2\pi r^2$ part is the top and the bottom.

$2\pi(7)(8) = 112\pi$ square centimeters.

3. Regular **462π square feet**

The sheet metal covers both bases and the curved surface: the whole surface area.

$\text{SA} = 2\pi(11)^2 + 2\pi(11)(10) = 242\pi + 220\pi = 462\pi$ square feet.

Comparison strip (2 items, C-GR.2.2-X) — used on a later week's re-check when this benchmark was *not* retaught

1. Regular **407π square inches**

With no top, the container has one base and the curved surface.

One base: $\pi(11)^2 = 121\pi$.

Curved surface: $2\pi(11)(13) = 286\pi$.

Total: $121\pi + 286\pi = 407\pi$ square inches.

2. Regular **188.40 square inches**

The label covers only the curved side, the $2\pi rh$ part of the formula; the $2\pi r^2$ part (the top and the bottom) is not included: $2\pi(3)(10) = 60\pi$.

With 3.14 for π: $60 \times 3.14 = 188.40$ square inches.

Reteach plan — C-GR.2.2

MA.7.GR.2.2 · Real-world cylinder surface area

Item that gave the most trouble (from both tally sheets):

What the mistake looks like:

Worked example for Tuesday (8–10 min):

Three practice items (one change at a time):

Decision line: Reteach on _____ Huddle: in person / paper Date: _____ Re-check given on _____

7th Grade Math · Re-check (3 items) C-GR.2.2-R

Benchmark: **MA.7.GR.2.2** · Real-world cylinder surface area Student code: _____ Calculator: yes

1. Surface area of a right circular cylinder: $\mathrm{SA} = 2\pi r^2 + 2\pi rh$

A cylindrical planter has no top. It has a radius of 5 inches and a height of 12 inches. The outside of the planter, including its bottom, will be painted.

What is the area that will be painted? Round your answer to the nearest hundredth. Use the π key on your calculator.

2. Surface area of a right circular cylinder: $\mathrm{SA} = 2\pi r^2 + 2\pi rh$

A cylindrical can has a radius of 7 centimeters and a height of 8 centimeters. A paper label wraps around the side of the can. It does not cover the top or the bottom.

What is the area of the label? Write your answer in terms of π.

3. Surface area of a right circular cylinder: $\mathrm{SA} = 2\pi r^2 + 2\pi rh$

A cylindrical water tank has a radius of 11 feet and a height of 10 feet. Its whole outside surface, including the top and the bottom, will be covered with sheet metal.

What is the area that the sheet metal must cover? Write your answer in terms of π.

Score: ____ / 3 → copy to your progress strip

7th Grade Math · Comparison items (2) C-GR.2.2-X

Benchmark: **MA.7.GR.2.2** · Real-world cylinder surface area Student code: _____ Calculator: yes

1. Surface area of a right circular cylinder: $\mathrm{SA} = 2\pi r^2 + 2\pi rh$

A cylindrical container is made of plastic. It has a radius of 11 inches and a height of 13 inches, and it has a bottom but no top.

What is the area of the plastic needed to make the container? Write your answer in terms of π.

2. Surface area of a right circular cylinder: $\mathrm{SA} = 2\pi r^2 + 2\pi rh$

A cylindrical oatmeal container has a radius of 3 inches and a height of 10 inches. A label wraps all the way around its curved side, with no overlap; the top and the bottom are not covered.

What is the area of the label? Round your answer to the nearest hundredth. Use 3.14 for π.

Score: ____ **/ 2** → **copy to your progress strip**

7th Grade Math · Comparison items (2) C-GR.2.2-X

Benchmark: **MA.7.GR.2.2** · Real-world cylinder surface area Student code: _____ Calculator: yes

1. Surface area of a right circular cylinder: $\mathrm{SA} = 2\pi r^2 + 2\pi rh$

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A cylindrical oatmeal container has a radius of 3 inches and a height of 10 inches. A label wraps all the way around its curved side, with no overlap; the top and the bottom are not covered.

What is the area of the label? Round your answer to the nearest hundredth. Use 3.14 for π.

Score: ____ **/ 2** → **copy to your progress strip**